Department of Electrical Engineering
विद्युत अभियांत्रिकी विभाग

Network Lab
नेटवर्क लैब

Experiment Number 2
प्रयोग संख्या 2

Verification of Superposition Theorem

सुपरपोजिशन प्रमेय का सत्यापन

MAULANA AZAD NATIONAL INSTITUTEOF TECHNOLOGY, BHOPAL

NETWORK LAB

EXPERIMENT NO.02

Aim: To verify the Superposition theorem.

Apparatus Required:

1. Trainer kit
2. Connecting cords

Theory:

The superposition theorem for electrical circuits states that for a linear system the response (voltage or current) in any branch of a bilateral linear circuit having more than one independent source equals the algebraic sum of the responses caused by each independent source acting alone, where all the other independent sources are replaced by their internal impedances. To ascertain the contribution of each individual source, all of the other sources first must be "turned off" (set to zero) by:

1. Replacing all other independent voltage sources with a short circuit (thereby eliminating difference of potential i.e. V=0; internal impedance of ideal voltage source is zero (short circuit)).

2. Replacing all other independent current sources with an open circuit (thereby eliminating current i.e. I=0; internal impedance of ideal current source is infinite (open circuit)).

This procedure is followed for each source in turn, then the resultant responses are added to determine the true operation of the circuit. The resultant circuit operation is the superposition of the various voltage and current sources.

The superposition theorem is very important in circuit analysis. It is used in converting any circuit into its Norton equivalent or Thevenin equivalent.

The theorem is applicable to linear networks (time varying or time invariant) consisting of independent sources, linear dependent sources, linear passive elements (resistors, inductors, capacitors) and linear transformers.

Another point that should be considered is that superposition only works for voltage and current but not power. In other words, the sum of the powers of each source with the other sources turned off is not the real consumed power. To calculate power, we should first use superposition to find both current and voltage of each linear element and then calculate the sum of the multiplied voltages and currents.

SUPERPOSITION THEOREM:

Statement: Superposition theorem states that "In any linear bilateral network containing two or more sources, the response in any element is equal to the algebraic sum of the responses caused by individual sources acting alone, while the other sources are non- operative i.e., while considering the effect of individual sources, other ideal voltage sources and ideal current sources in the network are replaced by short circuit and open circuit across their terminals".

Procedure:

1. Make the connections as shown in fig.1 and measure the current 'I'.

2. Short circuit E2 (assuming the internal resistance of E2 source to be zero) as shown in fig.2 and note down the current I, when only E1 is acting.

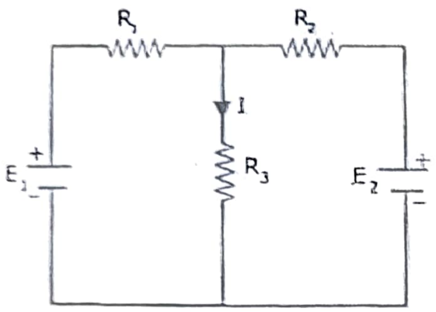
3. Note down the current in to millimetres and multi meter.

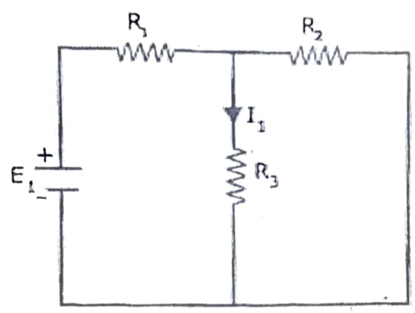
4. Short circuit E, (assuming the internal resistance of E, source to be zero) as shown in fig. 3 and note down the current 12 when only E2 is acting.

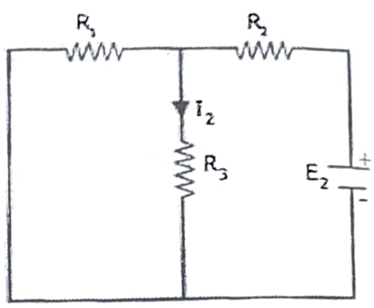
5. Note down the current in to milli meters and multi meter.

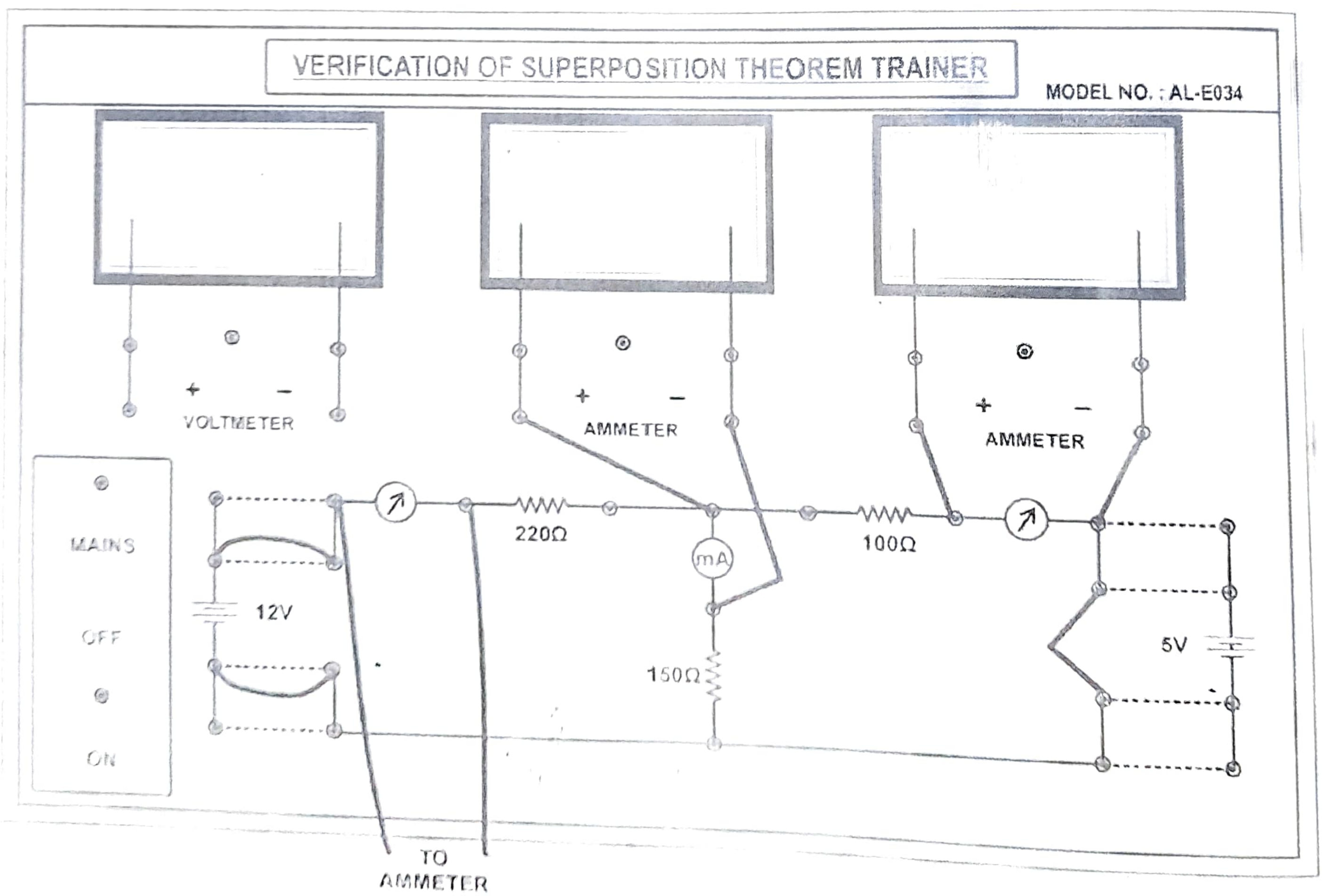
6. By superposition theorem $I = I_1+I_2$.

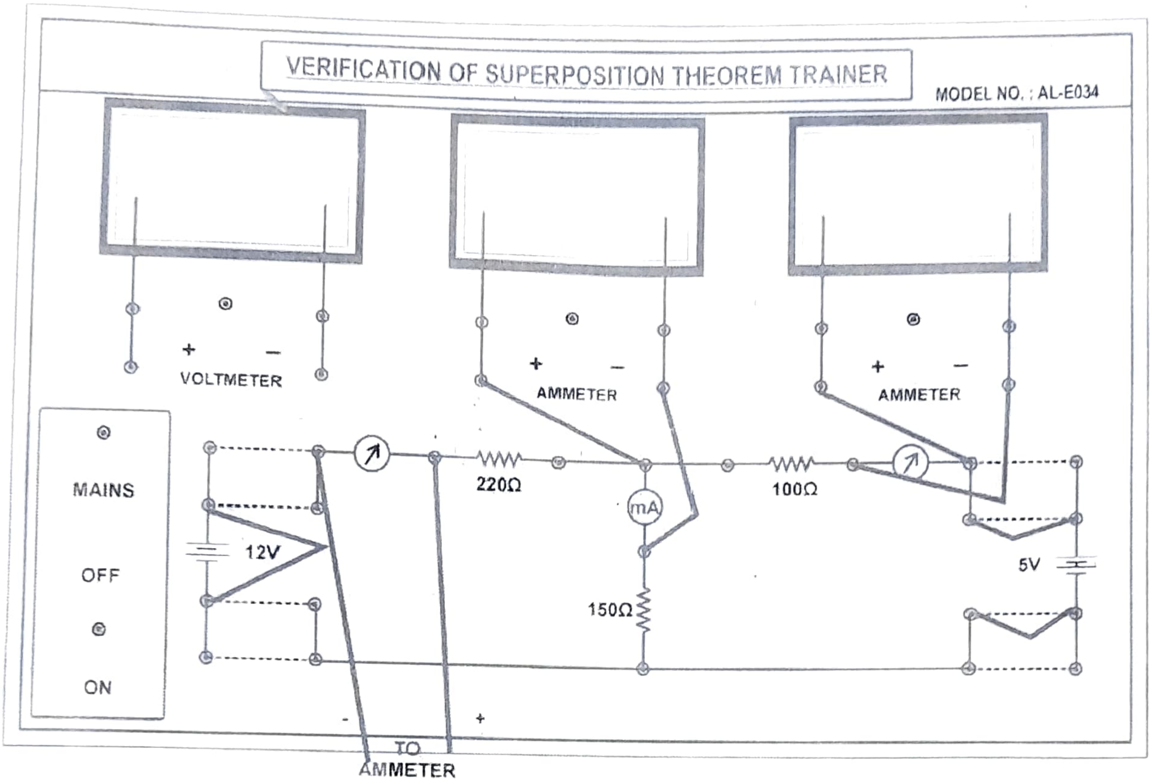
Connection diagram:











Observation table:

Parameters	Theoretical Values	Practical Values
I_1		
I_2		
I		

Result : If Practical and theoretical values meet near approximation. Then required result is verified.

Questions :

1. What is the superposition theorem?
2. Is the superposition theorem valid for AC circuits?
3. Is the superposition theorem applicable to power?
4. Can the superposition theorem be applied to non-linear circuits?
5. Why do we use the superposition theorem?
6. Limitations of Superposition Theorem?

ANS

1. Superposition theorem is a circuit analysis theorem that is used to solve the network where two or more sources are present and connected
2. The superposition theorem is valid for AC circuits.
3. The requisite of linearity indicates that the superposition theorem is only applicable to determine voltage and current, but not power. Power dissipation is a nonlinear function that does not algebraically add to an accurate total when only one source is considered at a time.
4. No, the superposition theorem can only be applied to non-linear circuits.
5. The superposition theorem is very important in circuit analysis because it converts a complex circuit into a Norton or Thevenin equivalent circuit.
6. The theorem does not apply to non-linear circuits. The requisite of linearity indicates that the superposition theorem is only applicable to determine voltage and current but not power. Power dissipation is a nonlinear function that does not algebraically add to an accurate total when only one source is considered at a time.

   The application of the superposition theorem requires two or more sources in the circuit.